# Operating Systems · 21CSC202J

Unit 1 · Session 5 · SLO 1 — Solved Worksheet

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## Unit â€“ 1 : Session â€“ 5 : SLO â€“ 1 | Exokernel and Layered Operating Systems â€” Word Reassembly

### I. Key Terms (reassembled)

• EXOKERNEL â€” a minimalist kernel that exposes hardware directly to applications through library
• operating systems; applications control resource management themselves.
• LAYERED STRUCTURE â€— the OS is divided into layers; each layer uses services only of the layer below
• it (layer 0 = hardware, top layer = user interface).
• IPC (Inter-Process Communication) â€— minimal service kept inside small kernels.
• LIBRARY OS â€— in exokernels, untrusted application libraries implement abstractions like files.
• MODULARITY â€— layered design allows debugging and verification layer by layer.

### II. Summary Answers

• Exokernel: purpose is maximum flexibility and performance by multiplexing hardware securely and
• letting applications manage resources via library OSes (e.g. MIT's Aegis/XN).
• Layered OS: architecture where functions are grouped into ordered layers; advantage is simplicity
• of construction/debugging; disadvantage is overhead of crossing layers and careful layer definition.